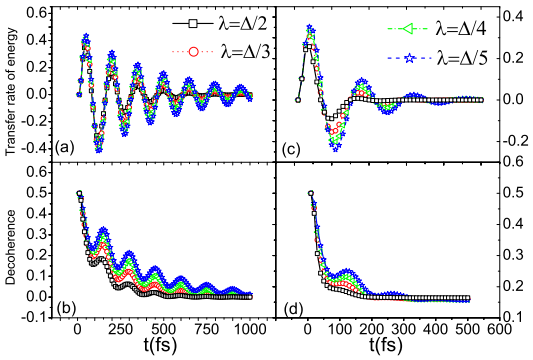

FIG. 2: (Color online) Time evolutions of the coherence [upper panels (a) and (c)] and transfer rate of the EET [lower panels (b) and (d)], in Markovian approximation $\delta k_{\max} = 0$ [left panels (a) and (b)], and in non-Markovian approximation $\delta k_{\max} = 3$ [right panels (c) and (d)], for different values of λ. The initial state is $|\psi_0\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$. The values of other parameters are the same as in Fig.1.

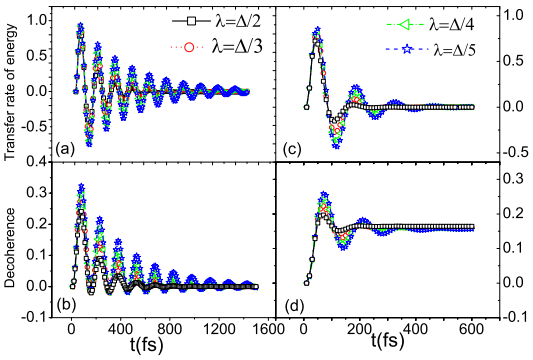

FIG. 3: (Color online) Time evolutions of the coherence and transfer rate of the EET in Markovian and non-Markovian approximations. The parameters in Fig.3 are similar to them in Fig.2 except the initial state $|\psi_0\rangle = |1\rangle$.

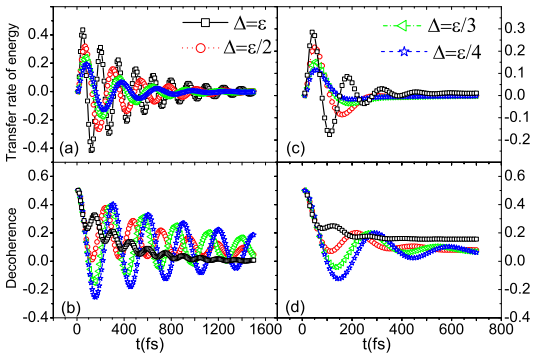

FIG. 4: (Color online) Time evolutions of the coherence [upper panels (a) and (c)] and transfer rate of the EET [lower panels (b) and (d)], in Markovian approximation $\delta k_{\max} = 0$ [left panels (a) and (b)], and in non-Markovian approximation $\delta k_{\max} = 3$ [right panels (c) and (d)], for different values of Δ. The initial state is $|\psi_0\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$. The values of other parameters are the same as in Fig.1.

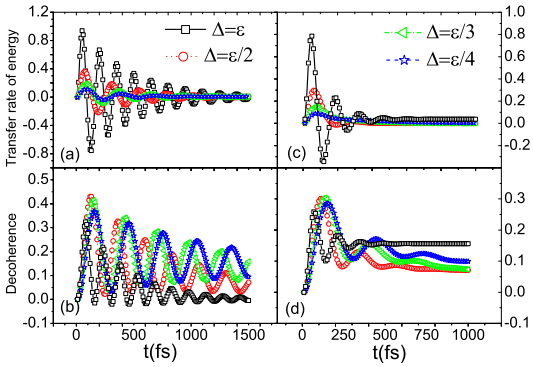

FIG. 5: (Color online) Time evolutions of the coherence and transfer rate of the EET in Markovian and non-Markovian approximations. The parameters in Fig.5 are similar to them in Fig.4 except the initial state $|\psi_0\rangle = |1\rangle$.

rate of energy k from the site 1 to the site 2 are oscillations, the coherent term $\rho_{12}(t)$ are also oscillations. The two kinds of evolution curves almost have the same decay times. It means that the system is transferring the energy in all of the coherence time. However, the maximum of k is nether at the maximal points nor at the minimal points of the curve $\rho_{12}(t)$. This means that the increase of coherence does not result in the increase of the transfer rate of the EET. Thus, we obtain the first conclusion in this paper that coherence helps the energy transfer through prolonging the transfer time rather than increase the transfer rate. As seen in the figures and tables that although the time of energy transfer is extended due to the increase of coherence its efficiency may decrease because the energy will be transferred between the site 1 and the site 2 back and forth. The breath-like transfer of energy in fact is ineffective. (2) From the Tables 1 and 2 we see that the environment really helps the energy transfer. It is clearly shown that an environment with stronger damping may stop the breath-like transfer of energy, which increase the transfer quality of the EET. In particular, as the non-Markovian effects are included the net area under the curve k, namely, the transfer quality of the EET is increased greatly in all cases. It means that the environment with non-Markovian effects will help the EET. This is the second conclusion in this paper. It is worthwhile to point out that as the initial state is the coherent superposition state the energy may be transferred from site 2 to 1 then the net area of the k curve is negative as shown at 410 fs in Table 2.